



Cambridge International AS & A Level

CANDIDATE
NAME
CENTRE
NUMBER

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CANDIDATE
NUMBER

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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

October/November 2024

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.



Find, in either order, the value of u and the value of H . [5]

[illegible]

- (a) Given that the particle just reaches O in the subsequent motion, find the value of d . [6]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for writing. The lines are evenly spaced across the entire page, providing a guide for letter height and placement. There is no text or other markings on the page.



[2]

[illegible]



(b) Given that $h = \frac{8}{3}a$, find the possible values of k . [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- $$\frac{v}{x} = \frac{3-t}{1+t}$$

(a) Find an expression for x in terms of t .

[4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



(b) Find the magnitude of F when $t = 3$.

[3]

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- As a result of the impact, P moves vertically upwards. The coefficient of restitution between P and the inclined plane is e .

(a) Show that $e \tan^2 \alpha = 1$. [4]

This image shows a full page of a document template designed for handwriting practice. It consists of multiple sets of three horizontal lines each, spanning the width of the page. Each set includes a solid top line, a dashed middle line, and a solid bottom line, providing a guide for letter height and placement. The entire page is otherwise blank, with no text or other markings.



(b) Find the value of e . [6]

[illegible]

[illegible]

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